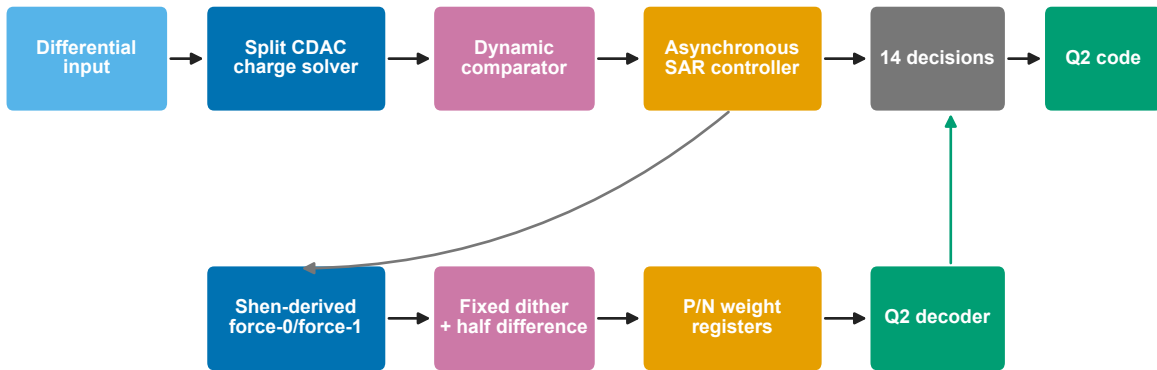


12-bit asynchronous SAR ADC Python behavioral model



Physical path: charge conservation + P/N mismatch. Calibration path: comparator-observable weights. Acceptance path: coherent FFT + static codebook audits.